

Fengbei Liu

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EDUCATION

Cornell Tech, Cornell University

Postdoctoral Researcher;

Supervisor: Prof. Mert R. Sabuncu

New York City, NY, US

Feb 2024 – Present

Australian Institute for Machine Learning (AIML)

Ph.D in Computer Vision and Medical Imaging;

Supervisor: Prof. Gustavo Carneiro, Prof. Mark Jenkinson

Adelaide, Australia

Apr 2020 – Oct 2023

University of Adelaide

Honors in Computer Science; (First Class)

Adelaide, Australia

Mar 2019 – Mar 2020

University of Adelaide

Bachelor in Computer Science;

Adelaide, Australia

Jul 2015 – Jul 2018

RESEARCH INTERESTS

Keywords: Computer Vision, Medical Image Analysis, Weakly-supervised Learning

My research interest is in computer vision and medical image analysis. I focus on developing weakly-supervised deep learning solutions and its application in medical imaging, including noisy label learning, semi-supervised learning, partial-label learning and active learning.

PUBLICATIONS

First-author Publications

Bridging Generative and Discriminative Noisy-Label Learning via Direction-Agnostic EM Formulation

Fengbei Liu, Chong Wang, Yuanhong Chen, Yuyuan Liu, Gustavo Carneiro

IEEE TPAMI 2026

RNED: Rotary Number Encoding and Decoding for Quantitative Medical VLM Analysis

Fengbei Liu, Sunwoo Kwak, Nusrat Binta Nizam, Ilan Richter, Ashley Beecy, Jayant K Raikhelkar, Deborah Estrin, Mert R. Sabuncu

CVPR 2026

HyperCT: Low-Rank Hypernet for Unified Chest CT Analysis

Fengbei Liu, Sunwoo Kwak, Hao Phung, Nusrat Binta Nizam, Ilan Richter, Nir Uriel, Hadar Averbuch-Elor, Deborah Estrin, Mert R. Sabuncu

MIDL 2026

BoMD: Bag of Multi-label Local Descriptors for Noisy Chest X-ray Classification

Yuanhong Chen, Fengbei Liu*, Hu Wang, Chong Wang, Yuyuan Liu, Yu Tian, Gustavo Carneiro*

ICCV 2023

Self-supervised Multi-class Pre-training for Unsupervised Anomaly Detection and Segmentation in Medical Images

Yu Tian, Fengbei Liu*, Guansong Pang, Yuanhong Chen, Yuyuan Liu, Johan W. Verjans, Rajvinder Singh, Gustavo Carneiro*

Medical Image Analysis 2023

NVUM: Non-volatile Unbiased Memory for Robust Medical Image Classification

Fengbei Liu, Yuanhong Chen, Yu Tian, Yuyuan Liu, Chong Wang, Vasileios Belagiannis, Gustavo Carneiro

MICCA 2022, Early Accept, featured in ComputerVisionNews

ACPL: Anti-curriculum Pseudolabelling for Semi-supervised Medical Image Classification

Fengbei Liu, Yu Tian*, Yuanhong Chen, Yuyuan Liu, Vasileios Belagiannis, Gustavo Carneiro*

CVPR 2022

Self-supervised Mean-teacher for Semi-supervised Chest X-ray Classification

Fengbei Liu, Yu Tian, Filipe R. Cordeiro, Vasileios Belagiannis, Ian Reid, Gustavo Carneiro

MICCAI-MLMI 2021

Self-supervised Depth estimation to Regularise Semantic Segmentation in Knee Arthroscopy
Fengbei Liu, Yaqub Jonmohamadi, Gabriel Maicas, Ajay K Pandey, Gustavo Carneiro
MICCAI 2020

Co-author Publications

BackSplit: The Importance of Sub-dividing the Background in Biomedical Lesion Segmentation
Rachit Saluja, Asli Cihangir, Ruining Deng, Johannes C. Paetzold, Fengbei Liu, Mert R. Sabuncu
CVPR 2026

From Failure to Feedback: Group Revision Unlocks Hard Cases in Object-Level Grounding
Yuyuan Liu, Yijing Ji, Anjie Le, Jiayuan Zhu, Jiazhen Pan, Can Peng, Jiajun Deng, Fengbei Liu, Junde Wu
CVPR 2026

Beyond Machine Interpretation: Learning from Expert Over-Reads Improves ECG Diagnosis
Sunwoo Kwak, Fengbei Liu, Nusrat Binta Nizam, Ilan Richter, Nir Uriel, Peter M. Okin, Mert R. Sabuncu
MIDL 2026

X-Cardia: Phenotype-Guided Cross-Modal Alignment for Opportunistic Cardiac Screening on Routine Chest CT
Nusrat Binta Nizam, Fengbei Liu, Sunwoo Kwak, Ilan Richter, Jayant K Raikhelkar, Ashley Beecy, Nir Uriel, Deborah Estrin, Mert R. Sabuncu
MIDL 2026, Oral

ShapKO: Shapley-Adaptive Modality Knockout for Robust Multimodal Learning
Nusrat Binta Nizam, Fengbei Liu, Sunwoo Kwak, Minh Nguyen, Ruining Deng, Mert R. Sabuncu
MICCAI 2026

An Artificial Intelligence Model to Detect Abnormal Ejection Fraction from Non-Contrast Chest Computed Tomography: The CT-LVEF study
Jayant Raikhelkar, Zilong Bai, Ashley N Beecy, Ilan Richter, Fengbei Liu, et al.
European Heart Journal-Digital Health 2026

Cardio Amyloid-AI: Advanced Multimodal Screening for Transthyretin Cardiac Amyloidosis in Severe Aortic Stenosis Patients
Nusrat Binta Nizam, Ashley N Beecy, Jacob W Groenendyk, Fengbei Liu, et al.
European Heart Journal-Digital Health 2026

Progressive Mining and Dynamic Distillation of Hierarchical Prototypes for Disease Classification and Localisation
Chong Wang, Fengbei Liu, Yuanhong Chen, Gustavo Carneiro
IEEE Journal of Biomedical and Health Informatics 2025

Cross- and Intra-image Prototypical Learning for Multi-label Disease Diagnosis and Interpretation
Chong Wang, Fengbei Liu, Yuanhong Chen, Helen Frazer, Gustavo Carneiro
IEEE Transactions on Medical Imaging 2025

Mixture of Gaussian-distributed Prototypes with Generative Modelling for Interpretable and Trustworthy Image Recognition
Chong Wang, Yuanhong Chen, Fengbei Liu, Yuyuan Liu, Davis J McCarthy, Helen Frazer, Gustavo Carneiro
IEEE Transactions on Pattern Analysis and Machine Intelligence 2025

Knockout: A Simple Way to Handle Missing Inputs
Minh Nguyen, Batuhan K. Karaman, Heejong Kim, Alan Q. Wang, Fengbei Liu, Mert R. Sabuncu
Transactions on Machine Learning Research 2025

Translation Consistent Semi-supervised Segmentation for 3D Medical Images
Yuyuan Liu, Yu Tian, Chong Wang, Yuanhong Chen, Fengbei Liu, Vasileios Belagiannis, Gustavo Carneiro
IEEE Transactions on Medical Imaging 2024

BRAIxDet: Learning to Detect Malignant Breast Lesion with Incomplete Annotations
Yuanhong Chen, Yuyuan Liu, Chong Wang, Michael Elliott, Chun Fung Kwok, Yu Tian, Fengbei Liu, Helen Frazer, Davis J McCarthy, Gustavo Carneiro
Medical Image Analysis 2024

Unraveling Instance Associations: A Closer Look for Audio-Visual Segmentation
Yuanhong Chen, Yuyuan Liu, Hu Wang, Fengbei Liu, Chong Wang, Helen Frazer, Gustavo Carneiro
CVPR 2024

Effective Segmentation of Post-Treatment Gliomas Using Simple Approaches: Artificial Sequence Generation and Ensemble Models

Heejong Kim, Léo Milecki, Mina C. Moghadam, **Fengbei Liu**, Minh Nguyen, Eric Qiu, Abhishek Thanki, Mert R. Sabuncu

MICCAI BraTS Challenge 2024, Oral Presentation

An Interpretable and Accurate Deep-learning Diagnosis Framework Modelled with Fully and Semi-supervised Reciprocal Learning

Chong Wang, Yuanhong Chen, **Fengbei Liu**, Michael Elliott, Chun Fung Kwok, Carlos Peña-Solorzano, Helen Frazer, Davis James McCarthy, Gustavo Carneiro

IEEE Transactions on Medical Imaging 2023

Learning Support and Trivial Prototypes for Interpretable Image Classification

Chong Wang, Yuyuan Liu, Yuanhong Chen, **Fengbei Liu**, Yu Tian, Davis J McCarthy, Helen Frazer, Gustavo Carneiro

ICCV 2023

Unsupervised anomaly detection in medical images with a memory-augmented multi-level cross-attentional masked autoencoder

Yu Tian, Guansong Pang, Yuyuan Liu, Chong Wang, Yuanhong Chen, **Fengbei Liu**, Rajvinder Singh, Johan W Verjans, Gustavo Carneiro

MICCAI-MLMI 2023

Knowledge Distillation to Ensemble Global and Interpretable Prototype-Based Mammogram Classification Models

Chong Wang, Yuanhong Chen, Yuyuan Liu, Yu Tian, **Fengbei Liu**, Davis J McCarthy, Michael Elliott, Helen Frazer, Gustavo Carneiro

MICCAI, 2022, Early Accept

Multi-view Local Co-occurrence and Global Consistency Learning Improve Mammogram Classification Generalisation

Yuanhong Chen, Hu Wang, Chong Wang, Yu Tian, **Fengbei Liu**, Yuyuan Liu, Michael Elliott, Davis J McCarthy, Helen Frazer, Gustavo Carneiro

MICCAI 2022, Early Accept

Contrastive Transformer-based Multiple Instance Learning for Weakly Supervised Polyp Frame Detection

Yu Tian, Guansong Pang, **Fengbei Liu**, Yuyuan Liu, Chong Wang, Yuanhong Chen, Johan W Verjans, Gustavo Carneiro

MICCAI 2022, Early Accept

Perturbed and Strict Mean Teachers for Semi-supervised Semantic Segmentation

Yuyuan Liu, Yu Tian, Yuanhong Chen, **Fengbei Liu**, Belagiannis Vasileios, Gustavo Carneiro

CVPR 2022

Pixel-wise Energy-biased Abstention Learning for Anomaly Segmentation on Complex Urban Driving Scenes

Yu Tian*, Yuyuan Liu*, Guansong Pang, **Fengbei Liu**, Yuanhong Chen, Gustavo Carneiro

ECCV 2022, Oral

Constrained Contrastive Distribution Learning for Unsupervised Anomaly Detection and Localisation in Medical Images

Yu Tian, Guansong Pang, **Fengbei Liu**, Yuanhong Chen, Seon Ho Shin, Johan W Verjans, Rajvinder Singh, Gustavo Carneiro

MICCAI 2021

3D Semantic Mapping from Arthroscopy using Out-of-distribution Pose and Depth and In-distribution Segmentation Training

Yaqub Jonmohamadi, Shahnewaz Ali, **Fengbei Liu**, Jonathan Roberts, Ross Crawford, Gustavo Carneiro, Ajay K Pandey

MICCAI 2021

Preprint/Under-Review

MAdam: Metric-Aware Multi-Objective Adam

Fengbei Liu, Rachit Saluja, Sunwoo Kwak, Ruibo Wang, Ruining Deng, Heejong Kim, Johannes C. Paetzold, Mert R. Sabuncu

Agentic Large-Language-Model Systems in Medicine: A Systematic Review and Taxonomy
*Abdul Mohaimen Al Radi, Xu Cao, Fanyang Yu, **Fengbei Liu**, Yu Tian, et al.*

Asymmetric Co-teaching with Multi-view Consensus for Noisy Label Learning
***Fengbei Liu**, Yuanhong Chen, Chong Wang, Yuyuan Liu, Gustavo Carneiro*

PROFESSIONAL ACTIVITIES

Conference

Reviewer: ICCV 2021/2023, MICCAI 2021, CVPR 2022/2023/2024, ECCV 2022/2024, BMVC 2022, NeurIPS 2023/2024, ICLR 2024, ICML 2024

Journal

Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

AWARDS & ACHIEVEMENTS

Ph.D Scholarship: University of Adelaide, Apr 2020

SKILLS

Programming: Python & Pytorch, Slurm, Docker & Kubernetes

Language: Mandarin (native), English (professional)

REFERENCES

Prof. Gustavo Carneiro
Professor of AI and Machine Learning, University of Surrey
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Prof. Vasileios Belagiannis
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Prof. Mert R. Sabuncu
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